

MATH 241 Spring 2026 Homework #10

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In this problem, we will review the ideas discussed in Section 5.1.

What are some quantities that cannot be modeled by a discrete random variable?

Continuous quantities variables like time, physical measurements, and rates/speeds. True or False? Given a random variable X , it is possible that X does not have a probability mass function. True Define what is meant by a *continuous* random variable. A random variable X is said to be continuous if its Cumulative Distribution Function (CDF), $F(x) = P(X \leq x)$, is continuous everywhere. More practically, X is continuous if there exists a non-negative function $f(x)$ (called the probability density function, or PDF) defined for all real numbers such that for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

What is the value of c ? Hint: Apply Humpty-Dumpty.

After using the integration from 0 to 2, and pulling out the c constant, we solve for the anti derivative and set $c * \frac{8}{3} = 1$ and got $c = 3/8$. Suppose that X is a continuous random variable whose probability

density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the cumulative distribution function of X . $F(x)$ is:

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{3}{4}x^2 - \frac{1}{4}x^3 & \text{if } 0 < x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X]$.

$$E[X] = \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2) dx = \frac{3}{8} \int_0^2 (4x^2 - 2x^3) dx$$

$$E[X] = \frac{3}{8} \left[\frac{4}{3}x^3 - \frac{1}{2}x^4 \right]_0^2$$

$$E[X] = \frac{3}{8} \left(\left[\frac{4}{3}(8) - \frac{1}{2}(16) \right] - 0 \right) = \frac{3}{8} \left(\frac{32}{3} - 8 \right)$$

$$E[X] = \frac{3}{8} \left(\frac{32}{3} - \frac{24}{3} \right) = \frac{3}{8} \left(\frac{8}{3} \right) = 1$$

Answer: $E[X] = 1$ Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} c(4x - 2x^2) & \text{if } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X^2]$.

$$E[X^2] = \int_0^2 x^2 \cdot \frac{3}{8}(4x - 2x^2) dx = \frac{3}{8} \int_0^2 (4x^3 - 2x^4) dx$$

$$E[X^2] = \frac{3}{8} \left[x^4 - \frac{2}{5}x^5 \right]_0$$

$$E[X^2] = \frac{3}{8} \left(16 - \frac{2}{5}(32) \right) = \frac{3}{8} \left(16 - \frac{64}{5} \right)$$

$$E[X^2] = \frac{3}{8} \left(\frac{80}{5} - \frac{64}{5} \right) = \frac{3}{8} \left(\frac{16}{5} \right) = \frac{6}{5} \text{ (or 1.2)}$$

Answer: $E[X^2] = \frac{6}{5}$ Suppose that a random variable X measures the lifetime of a certain type of electronic device in hours. Suppose its probability density function is given by:

$$f(x) = \begin{cases} \frac{10}{x^2}, & \text{if } x > 10 \\ 0, & \text{otherwise} \end{cases}$$

What is the probability that of 6 such types of devices, at least 3 will function for at least 15 hours? (You may assume that the devices operate independently of one another). You may leave your answer in terms of a calculator command. $1 - \text{binomcdf}(6, 2/3, 2)$ is the answer.

These exercises will review the king of all random variables – the Normal random variable.

What is the density of the normal random variable?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

What is the support of the normal random variable? The support is all real numbers: $(-\infty, \infty)$. What is the cumulative distribution function of the normal random variable?

The CDF does not have a closed-form algebraic solution and is expressed as an integral:

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt$$

If $X \sim \mathcal{N}(\mu, \sigma^2)$ then what is $E[X]$?

$$E[X] = \mu$$

If $X \sim \mathcal{N}(\mu, \sigma^2)$ then what is $\text{Var}[X]$?

$$\text{Var}[X] = \sigma^2$$

What is the standard normal random variable? It is a special normal random variable with a mean of 0 and a variance (and standard deviation) of 1. It is typically denoted by the letter Z , where $Z \sim N(0, 1)$. What is the density of the standard normal random variable? Use the appropriate letter/notation.

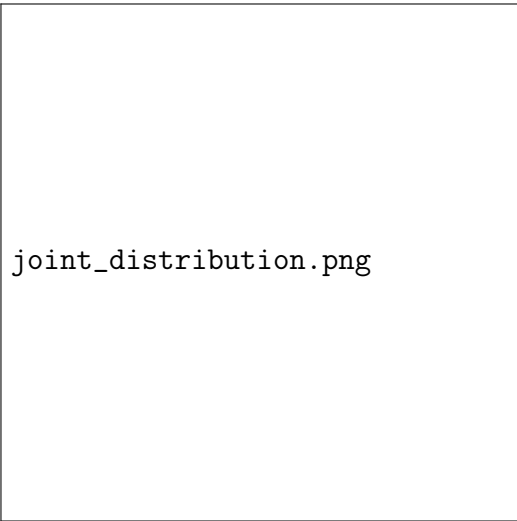
$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

What is the cumulative distribution function of the standard normal random variable? Use the appropriate letter/notation.

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-\frac{t^2}{2}} dt$$

Suppose $Z \sim \mathcal{N}(0, 1)$. Find $P(Z < 2.41)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Answer: $\Phi(2.41)$ or $\text{normalcdf}(-\infty, 2.41, 0, 1) \approx 0.9920$ Suppose $Z \sim \mathcal{N}(0, 1)$. Find $P(Z > 2.41)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Answer: $1 - \Phi(2.41)$ or $\text{normalcdf}(2.41, \infty, 0, 1) \approx 0.0080$ Suppose $Z \sim \mathcal{N}(0, 1)$. Find $P(Z = 2.41)$. Answer: 0 (The probability of a continuous random variable taking any exact value is always 0). Suppose $X \sim \mathcal{N}(241, 17)$. Find $P(X > 231)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Answer: $\text{normalcdf}(231, \infty, 241, \sqrt{17}) \approx 0.9924$ Suppose $X \sim \mathcal{N}(241, 17)$. Find $P(X < 231)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Answer: $\text{normalcdf}(-\infty, 231, 241, \sqrt{17}) \approx 0.0076$

These exercises are designed to review the first half of Section 5.4. Suppose the joint mass function of X and Y is given by:



joint_distribution.png

Define what is meant by a joint mass function (jmf). A joint probability mass function $p(x, y) = P(X = x, Y = y)$ is a function that gives the probability that two discrete random variables, X and Y , simultaneously take on specific values x and y . It must satisfy $p(x, y) \geq 0$ and $\sum_x \sum_y p(x, y) = 1$. Find $P(X \leq 2, Y \leq 2)$.

$$P(X \leq 2, Y \leq 2) = \sum_{x \leq 2} \sum_{y \leq 2} p(x, y)$$

Find $f_X(2)$. This represents the marginal probability that $X = 2$. Find the row where $X = 2$ and sum all the values across all columns (Y values) in that row. Find $f_Y(4)$. This represents the marginal probability that $Y = 4$. Find the column where $Y = 4$ and sum all the values down all rows (X values) in that column. Find $f_X(x)$. $f_X(x) = \sum_{all\ y} p(x, y)$ (The row totals) Find $f_Y(y)$. $f_Y(y) = \sum_{all\ x} p(x, y)$ (The column totals) What does it mean for two random variables, X and Y , to be independent? Two discrete random variables are independent if and only if their joint PMF is equal to the product of their marginal PMFs for all possible pairs of (x, y) :

$$p(x, y) = f_X(x) \cdot f_Y(y)$$

For the joint probability mass function specified above, is X and Y independent? Explain why or why not. To check: Pick a single cell in

your matrix, say $(x = 1, y = 1)$.

Multiply its row total $(f_X(1))$ by its column total $(f_Y(1))$.

If $f_X(1) \cdot f_Y(1) = p(1, 1)$, check the other cells.

If it holds true for every single cell, they are independent.

If even a single cell fails this multiplication equality, write:

"No, because $p(x, y) \neq f_X(x) \cdot f_Y(y)$ for cell (x, y) ."

In this problem, we will review the last part of Section 5.4.

Explain what is meant by a convolution of random variables. A convolution is a mathematical operation used to find the probability distribution of the sum of two or more independent random variables ($T = X + Y$). For discrete variables, it involves summing over all possible mutually exclusive ways the individual variables can add up to a target value t . Suppose $X_1, X_2 \sim p$. Let $T = X_1 + X_2$. What is the support of T ? Support: $\{2, 3, 4, 5, \dots\}$ Find $P(T = 2)$. The only way to get a sum of 2 is if $X_1 = 1$ and $X_2 = 1$:

$$P(T = 2) = P(X_1 = 1) \cdot P(X_2 = 1) = p \cdot p = p^2$$

Find $P(T = 3)$. Ways to sum to 3: $(1, 2)$ or $(2, 1)$.

$$P(T = 3) = P(X_1 = 1, X_2 = 2) + P(X_1 = 2, X_2 = 1) = p(qp) + (qp)p = 2p^2q \quad (\text{where } q = 1 - p)$$

Find $P(T = 4)$. Ways to sum to 4: $(1, 3)$, $(2, 2)$, or $(3, 1)$.

$$P(T = 4) = p(q^2p) + (qp)(qp) + (q^2p)p = 3p^2q^2$$

Based off the last three computations, can you guess the probability mass function of T ? You do not have to mathematically prove your answer. (If you are curious, the actual proof requires induction and the general convolution formula which you are not required to know). Following the pattern $(1p^2, 2p^2q^1, 3p^2q^2)$:

$$P(T = t) = (t - 1)p^2(1 - p)^{t-2} \quad \text{for } t = 2, 3, 4, \dots$$

Here is the last problem for the semester. Congratulations on making it this far! These exercises are designed to review Section 5.5.

Without simply quoting the formal statement of the Central Limit Theorem, can you explain what the Central Limit Theorem states? The

CLT states that if you take a large sample size ($n \geq 100$) of independent, identically distributed random variables from any underlying population distribution (even skewed or weird ones) with a finite mean μ and variance σ^2 , the distribution of the sample sum (or sample mean) will naturally form a Normal Distribution shape. Suppose the random variables, $X_1, X_2, \dots, X_{100}$ are independent and all have the following density:

$$f(x) = \begin{cases} \frac{x^2}{72} & \text{if } 0 < x < 6 \\ 0, & \text{otherwise} \end{cases}$$

If $Y = X_1 + X_2 + \dots + X_{100}$, then find/approximate $P(439 \leq Y \leq 461)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places. Step 1: Find the mean μ and variance σ^2 of a single variable X .

$$\mu = E[X] = \int_0^6 x \cdot \frac{x^2}{72} dx = \frac{1}{72} \left[\frac{x^4}{4} \right]_0^6 = \frac{1296}{288} = 4.5$$

$$E[X^2] = \int_0^6 x^2 \cdot \frac{x^2}{72} dx = \frac{1}{72} \left[\frac{x^5}{5} \right]_0^6 = \frac{7776}{360} = 21.6$$

$$\sigma^2 = E[X^2] - (E[X])^2 = 21.6 - (4.5)^2 = 21.6 - 20.25 = 1.35$$

Step 2: Find the expected value and variance for the sum Y .

$$\mu_Y = n\mu = 100(4.5) = 450$$

$$\sigma_Y^2 = n\sigma^2 = 100(1.35) = 135 \implies \sigma_Y = \sqrt{135}$$

By CLT, $Y \approx N(450, 135)$. Standardizing gives:

$$P(439 \leq Y \leq 461) \approx \text{normalcdf}(439, 461, 450, \sqrt{135})$$

$$\text{Answer: } \Phi\left(\frac{11}{\sqrt{135}}\right) - \Phi\left(\frac{-11}{\sqrt{135}}\right) \approx 0.6551$$

Suppose the random variables $X_1, X_2, \dots, X_{36}$ are independent and all have the density specified below.

$$f(x) = \begin{cases} \frac{1}{20}x^3 & \text{if } 1 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

Find/approximate $P(2 \leq \bar{X}_{36} \leq 2.5)$. You may leave your answer in terms of an integral, in terms of Φ , in terms of a calculator command, or in terms of an actual number rounded to four decimal places.

$$\begin{aligned} P(2 \leq X_{36} \leq 2.5) &= \int_2^{2.5} \frac{1}{20x^3} dx = \frac{1}{20} \int_2^{2.5} x^{-3} dx \\ &= \frac{1}{20} \left[-\frac{1}{2x^2} \right]_2^{2.5} = -\frac{1}{40} \left[\frac{1}{(2.5)^2} - \frac{1}{2^2} \right] = -\frac{1}{40} \left[\frac{1}{6.25} - \frac{1}{4} \right] \\ &= -\frac{1}{40} [0.16 - 0.25] = -\frac{1}{40} [-0.09] = \frac{0.09}{40} = 0.00225 \end{aligned}$$

Answer: 0.0023